

Security incident report

Section 1: Identify the network protocol involved in the incident

HTTP (Hypertext Transfer Protocol) was the primary network protocol involved in the incident. HTTP was used to request and retrieve the webpage, and the malicious file was delivered to users through HTTP traffic at the application layer.

Section 2: Document the incident

The incident was discovered after several customers contacted the helpdesk and reported being prompted to download a file while visiting yummyrecipesforme.com. After running the file, customers reported that their computers began operating more slowly. The website owner also discovered that they were unable to access the administrative account.

The former employee executed a brute-force attack to gain access to the web host administrative account by repeatedly guessing known default passwords. Once access was gained, the hacker changed the administrator password, locking the account owner out. The hacker then modified the website's source code and inserted malicious JavaScript that prompted visitors to download an executable file disguised as a browser update. After the file was downloaded and executed, the visitor's browser was redirected from yummyrecipesforme.com to greatrecipesforme.com, which contained the malware.

The tcpdump log supports the investigation by showing DNS and HTTP traffic associated with both websites. The computer first used DNS to resolve yummyrecipesforme.com to 203.0.113.22 and then requested the webpage using HTTP. Later, the computer performed another DNS lookup for greatrecipesforme.com, which resolved to 192.0.2.17, and then established a connection and sent an HTTP request to that website.

Section 3: Recommend one remediation for brute force attacks

Limiting failed login attempts would help prevent brute-force attacks by temporarily locking the account after a certain number of unsuccessful login attempts. This would prevent an attacker from continuously guessing passwords.